Efficiency of Algorithms
Space Complexity



Thareendhra Keerthi Wijayasiriwardhane

Department of Industrial Management
Faculty of Science
University of Kelaniya
thareen@kln.ac.lk





Learning Outcomes

- On completion of this session, you should be able to
  - Define the space complexity of an algorithm
  - Explain the importance of the analysis of space complexity
  - List the components of space complexity
  - Identify the factors on which the components of space complexity depend
  - Distinguish between sensitive and insensitive factors affecting space complexity
  - Analyze the space complexity of a given algorithm

Space Complexity

- Amount of memory that a program written with a particular algorithm needs to run to its completion
- Why are we interested?
  - Would like to know whether the available memory is enough to run the program written with our algorithm
  - Will prefer the program that consumes less memory when there are several algorithms for a given problem
  - Could estimate the largest problem that our algorithm can solve against the memory available

Space Complexity

- Components
  - Instruction space
  - Data space
  - Environment stack space

Instruction Space

- Memory needed to load the compiled version of a program (i.e. instructions)
- Depends on
  - Compiler used
  - Compiler options used
  - Architecture of the target computer

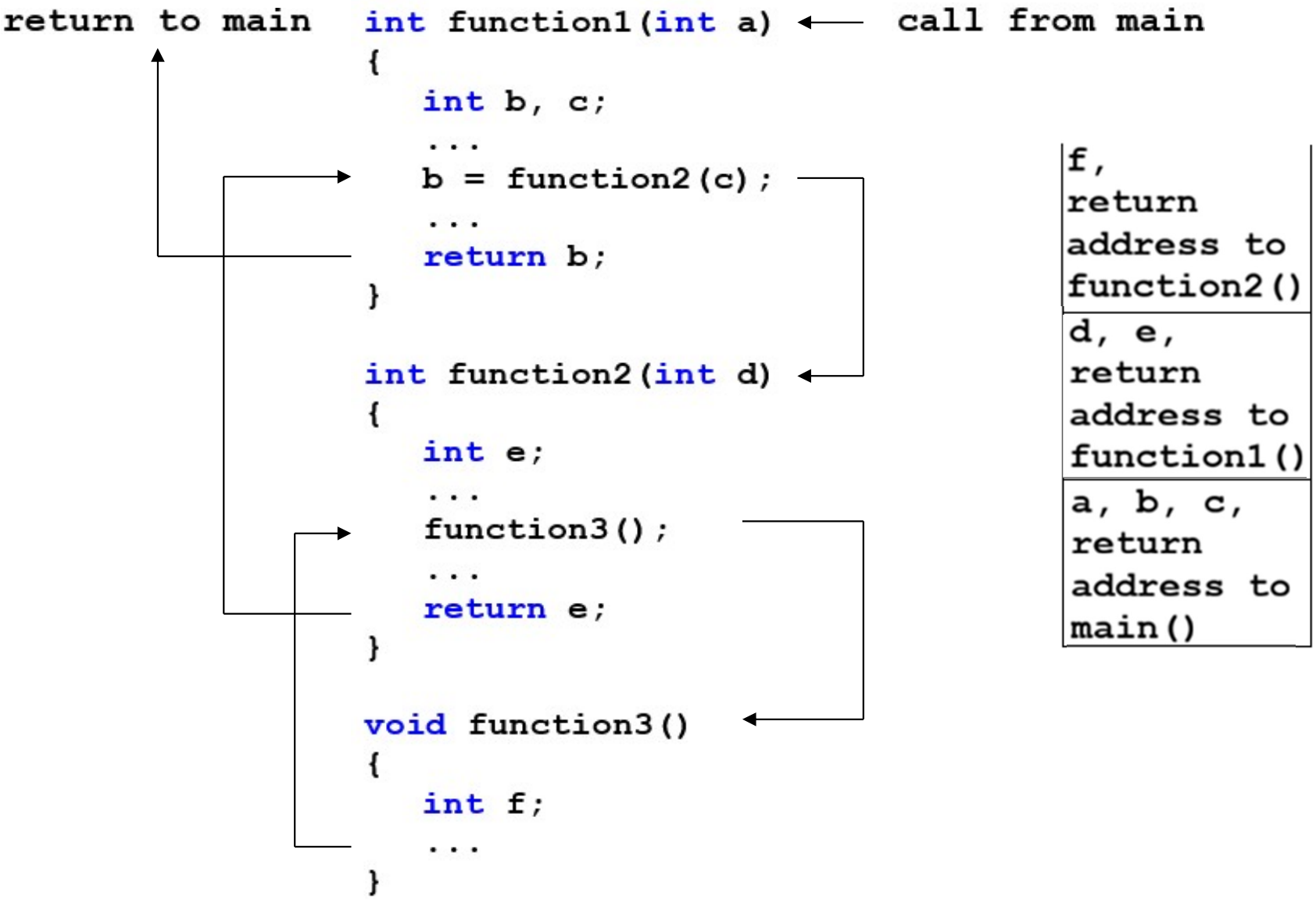
Data Space

- Memory needed to hold all the constants and variables of a program
- Depends on data types and the number of
  - Named constants used
  - Primitive variables used
  - Structured variables used

Environment Stack Space

- Memory needed for the environment stack
- What's environment stack?
  - A data structure that keeps information needed to resume execution of partially completed functions
  - Example
    If function1 calls function2, environment stack keeps a pointer to the function2's calling instruction in function1 (return address) to resume the execution of function1 when function2 returns

Environment Stack



Factors of Space Complexity

- Some factors are not known at the time of algorithm development
- Insensitive factors
  Relatively insensitive to the problem being solved
  - Instruction space contribution
  - Data space contribution from literal constants, named constants and primitive variables
  - Environment stack space contribution from non-recursive functions

Factors of Space Complexity

- Sensitive factors

  Sensitive to the problem being solved
  - Data space contribution from structured variables and dynamic memory allocations
    - Example - a program that sorts an array of n integers
  - Environment stack space contribution from recursive functions
    - Example - a function that recursively computes the factorial value of a given n
  - Space complexity is a function of n (instance characteristic)

Space Complexity

- Space complexity of algorithm A

  $$S(n) = c + S_A(n)$$

  where c is insensitive (constant) space contribution and S_A is the sensitive space contribution which is a function of instance characteristic n

Performance Analysis

- Example 1

```
ALGORITHM sum(REF a[] INTEGER, n INTEGER)
   returns sum of integers in array
1  s ← 0
2  i ← 0
3  LOOP (i < n)
4     s ← s + a[i]
5     i ← i + 1
6  END LOOP
7  RETURN s
END sum
```

Performance Analysis

- Example 2

```
ALGORITHM sum(REF a[] INTEGER, n INTEGER)
   returns sum of integers in array
1  IF (n > 0)
2     RETURN Sum(a, n - 1) + a[n - 1]
3  ELSE
4     RETURN 0
5  END IF
END sum
```

Questions